

Advanced modular battery management system for next generation of research platforms

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Abstract— The Battery Management System (BMS) is implemented as a cost-oriented design to monitor and protect the battery cells under their Safe Operation Area (SOA) and is structured in different logical blocks. Depending on the specific design, feature content and tailoring of the system, location of the features and software units may vary from design to design. New physical engineering units established with state-of-the-art sensing methods, cell balancing and reconfiguration technologies and increased performance of the battery cells within their SOA are currently under research. Wireless communication ensures modularity and scalability of the design and cloud connectivity enables digital twin and eases 2nd life reuse, avoiding unnecessary direct recycling. A fully scalable modular BMS for research purposes is a need in the research community, to this end a new intelligent Module Control Unit (iMCU) system is proposed. This study defines the iMCU and its framework to set the first milestone of a series of research studies analyzing different technology trends in the area of battery electronics and their application with the iMCU.

Index Terms — Modular BMS, wireless communication, cell balancing, sensing technologies

I. NOMENCLATURE

BMS – Battery Management Systems
BJB – Battery Junction Box
BMU – Battery Management Unit
PMU – Pack Management Unit
MMU – Module Management Unit
CMU – Cell Management Unit
SOA – Safe Operation Area
SOC – State of Charge
KPI – Key Performance Indicator
C2C – Cell to cell
C2P – Cell to pack
C2P2C – Cell to pack to cell
P2C – Pack to cell

II. INTRODUCTION

Traditional BMS architectures are designed in a centralized manner consisting in a central BMU controller, the BJB connected to the powertrain and the

CMUs monitoring the battery modules, constructed with sets of 8 up to 18 series connected cells, each cell may incorporate one or more cells connected in parallel, depending on cell capacity and chemistry selected [1]–[4]. On these traditional architectures, all computational resources are centralized in a master microcontroller, implementing all software algorithms in order to ensure a SOA of the battery cells. This master control drives the BJB, which controls the battery contactors, the pre-charge circuitry, and measures the battery input/output current as well as the battery string voltage before and after the contactors. For safe operation, insulation monitoring of battery pack and interlock and crash detection are also monitored centrally.

Correspondingly, CMU does not include any intelligence and performs only sensing of the battery cells (V, T) and implement passive balancing circuitries for equalization of the cells SOC.

Traditionally, communications among the BMU, BJB and CMU are based in CAN interface, however privative isoSPI [5]–[8] or Daisy Chain [9]–[11] communication buses, depending on the Integrated Circuit (IC) supplier selected for its application are present in current designs.

Even though such architectures present a very mature level of industrialization and are low-cost approach solutions in the automotive mass market, they offer very low options for real time scalability, modularity, refurbishing possibilities, and limit drastically the options of the battery pack designer during the early stages of the battery electronics design. Scalability for 800 V application with such IC technology presents a challenge on the real time monitoring of the battery cells. The large amount of data transfers through the wired network affects the periodicity and synchronicity of the measurements. With a high number of cells (around 160-192 cells to create a 800 V system), and consequently a corresponding high number of ICs, it is introduced a miss-synchronization and time delay in data transfer in the range of few milliseconds [10], [12]–[14].

Additionally, such applications are very limited in terms of accessing the data and status of the battery cells, becoming a challenge for third parties that attempt to reuse the EV battery packs in a stationary application or when executing a full recycle of the system in a closed LCA approach. Therefore, for 2nd life application, it is often needed a partial dismantling of the battery modules or even a cell re-sorting process in a full disassembly

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approach, which require characterization processes of the cell/module status, increasing the cost and potential time to market of these solutions [15]–[28].

Given that the BMS technology must be ready to address those issues and to incorporate the next generation of advances in battery electronics (such as innovative sensors and self-healing mechanisms) a new approach for a decentralized BMS is proposed in this work. This paper is structured as follows, in Section III the future BMS challenges are addressed, Section IV describes the iMCU and in Section V the current status of its implementation is overviewed. Section VI summarizes the conclusions of this work.

III. CHALLENGES ON FUTURE BMS ELECTRONICS

Feature content of the BMS can be structured logically in three different layers, similarly as [29], [30], depending on the hierarchy and information exchange (Fig. 1). These are:

- Decision layer: all SoX estimators, charging control, thermal management, balancing strategy (passive or active), reconfiguration strategy (if any).
- Monitoring layer: isolation monitoring, supervision, Protection Earthing (PE) monitoring, crash reaction, BMS off timer, SOA diagnostics and supervision, cybersecurity application over the different communication interfaces
- Sensing / Actuation / Communication layer: all measurements, independently of the CMU, MMU, BMU, PMU levels, such as: voltage, current, temperature, coolant, temperature, pressure, gas, humidity, isolation, interlock; as well all actuators: balancing / reconfiguration switches, isolation actuation, communication interfaces.

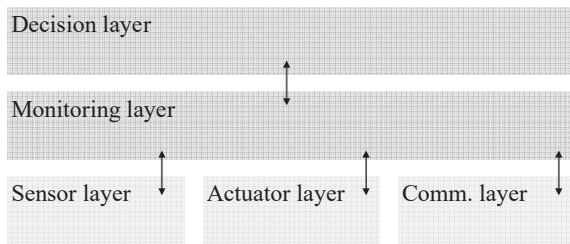


Fig. 1 Different layers of application software in a BMS environment.

Within these layers future innovations will be embedded. A summary of the major relevant trends is presented in the following subsections.

A. Modularity, scalability, and decentralization

Different studies over the recent years have focused on the applications of technologies in BMS targeting modularity and decentralization. Researchers have designed different feature applications either at CMU, MMU, BMU or PMU levels [31]–[36]. A proper scalable BMS shall be capable of acting as any of the roles

presented within a battery pack. Depending on the modularity needs, certain features shall be intelligently relocated to ensure SOA of the battery cells. This requires, depending on the granularity desired in the system, the introduction of calculation resources at CMU level.

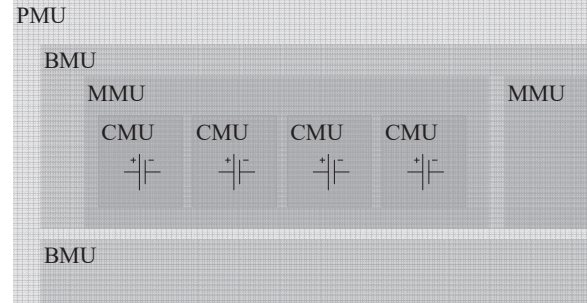


Fig. 2 Hierarchical distribution of calculation resources in battery systems.

Besides modularity or scalability needs, depending on the placement of the calculation resources over the different levels of abstraction existing in a BMS it is possible to introduce different levels of decentralization penetration. For instance, introducing parallel redundancy of monitoring of a specific battery cell to take over the feature content of a particular calculation resource, when this primary unit fails. Additionally, when the BMS does include digital twin integration, improved calculation for increased robustness can be performed in cloud, enabling edge calculation services.

Most of the decision layer features included in a BMS could be improved by application of decentralization and / or digital twin approach. In this sense, some of the other layers could be also transferred and relocated into cloud. This robustness improvement shall but not reduce the safety operation of the battery cells nor bring additional risks into the system by introducing additional software.

B. Application of new sensing technologies for advance modelling

Besides traditional measurements at cell level, like cell voltage and temperature; and usual measurements at battery module or pack level, as e.g., pack voltage, current through the pack, which are the basis for the current industrial BMS, upcoming additional technologies for additional sensors at sensing layer level are widely studied in the research field to add intelligence to the BMS. This trending topic of research serves to improve the performance of the BMS to extend their SOA, as well as some of these physical engineering units gain more importance for the new chemistries investigated, which help to understand in better way the behavior of the chemical process.

In

Table I a distribution of the new sensing technologies by ubication of the new sensor is shown, follow by a brief explanation of the new physical engineering unit at the sensing layer and their corresponding monitoring layer [37]–[41].

TABLE I
NEW TYPE OF SENSORS AND POSITIONING WITHIN THE BATTERY PACK

Sensor type	Position inside the battery			
	In cell	Over Cell	Module	Pack
Pressure	X	X	X	X
Gas			X	X
EIS		X	X	
Strain Gauge		X	X	
Electrode Volt.	X			
Internal Temperature	X			
Humidity			X	X

- Pressure sensing: an increase of the pressure could be detected inside the cell with a sensing device located inside or over the cell, with the purpose of detecting an increasing pressure within the vessel of the cell. By the case the module or pack is sealed, an increase of the pressure would reflect already an upcoming venting effect of any of the cells located.
- Detection of gas mixture is placed at module or pack level; it aims to analyze the gas mixture and detect abnormal concentration of certain gases that may indicate a potential failure of the battery.
- Electrochemical Impedance Spectroscopy (EIS) can be performed in different ways to characterize the impedance of a cell individually or a subset of cells, commonly a module. Different approaches to get the real and imaginary values of the cell impedance over frequency have been investigated, such as individual current generators parallel to each cell, use of DCDC converters as part of the balancing strategy or reusing the charging current for rough estimation of the impedance.
- Strain gauge to detect physical deformation of the pouch as well as rigid prismatic or cylindrical cell, nevertheless measurements at module level are also proposed to reduce the number of sensors and signals to be analyzed.
- Reference Electrode for voltage measurement for proper characterization of the reactions at the cathode and anode sides.
- Internal temperature sensor to achieve an accurate value of the cell core temperature, avoiding the effects of the insulation, encapsulation and sealing of the cell.
- Humidity, to detect leakages of the electrolyte in liquid electrolyte battery cells, which can be measured at the vessel of the module or pack, depending on the specific sealing of the design.

C. Active Balancing and Cell Reconfiguration as extended actuators in battery pack

Passive balancing strategies are cost oriented and

extended solution in industrialization phase since the need of equalization of battery cells is required. This solution is not effective and might be deprecated over the next years due to diverse factors,

Therefore, the introduction of loss-reduced battery equalization technologies, such as active balancing has been gaining research interest over the past years, as they could become an appropriate solution for:

- 1) Application of new chemistries which introduce a high production variability of capacity (e.g. cells manufactured with recycled materials), but are really cost oriented.
- 2) Legislation changes towards increase of the overall efficiency chain of the energy conversion, setting efficiency targets and absolute loss of energy values in the battery charging and discharging processes.
- 3) Extension of the RUL of the battery pack by reducing the average DoD over the lifetime
- 4) Qualification of the battery pack for a 2nd life application with a partial dismantling strategy.

Active balancing hardware implementations can be organized within two different parameters [14], [42]–[46]:

- 1) Configuration of the power components and topology used for the exchange of energy between the cells (capacitor, inductances, DCDC w/o galvanic isolation, DCDC topologies.
- 2) Configuration of the energy flow between the cells (few volts), the module typically < 60 V), the battery pack (> 300 V) or additional low voltage supply (typically a 12 V net or an auxiliar capacitance of such voltage level).

Table II approaches an overview of hardware topology application versus the energy flow of the balancing strategy.

TABLE II
ANALYSIS OF THE ACTIVE BALANCING ARCHITECTURES IN REGARDS OF THE ENERGY EXCHANGE FLOW OF THE CELLS

Hardware	Energy Flow				
	Direct C2C	Near C2C	P2C	C2P	C2P2C
Capacitor					
Switched		X			
Single Switched	X				X
Double-Tiered Switched	X				
Inductor					
Single Inductor	X			X	X
Coupled Inductor	X			X	
Transformer					

Hardware	Energy Flow				
	Direct C2C	Near C2C	P2C	C2P	C2P2C
Single windings			X	X	X
Multi windings			X	X	X
Multiple			X	X	X
DCDC converter					
Buck/Boost		X		X	
Cuk		X			
Ramp		X			
Full bridge		X			
Quasi resonant		X			
Flyback			X	X	X

Even though active balancing strategies have a strong presence in the research arena, reaching industrialization phase in the near time is not expected.

Besides active balancing strategies, another possibility for instance is reconfiguration of the battery string. With the introduction of additional switches in the power and auxiliary paths, surrounding the cell, it is aimed to modify series/parallel connection and/or by-pass malfunctioning cells.

This additional freedom grade to the battery pack in its configuration has been proposed in the literature [41], [47]–[53]. However, despite the different approaches investigated over the last years, the Technology Readiness Level (TRL) is still at prototype phase and not being considered for market introduction. In Figure 3 hardware implementations of the reconfiguration switches for battery cell application are presented. Here, the concept is presented at cell level, but each of the approaches is scalable at module or pack level, depending on the power rates and the industrial application.

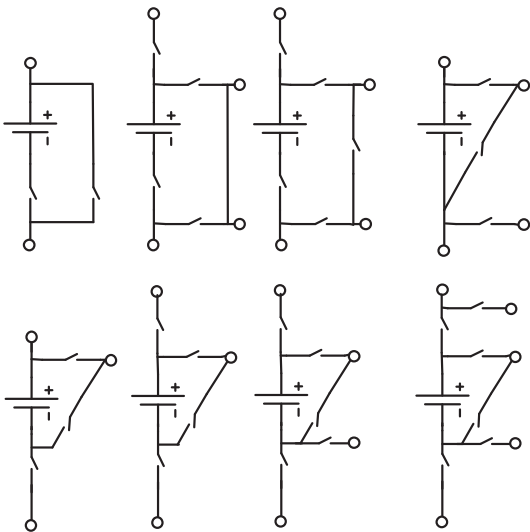


Fig. 3 Technology approaches of reconfiguration topologies of ESS.

D. Wireless, Connectivity

Wireless communication applied to BMS is a technology entering the mass market industrialization area. Several OEMs are introducing wireless information exchange between the BMU and the CMU/MMU, either with classical partnerships with TIER-1 component suppliers or directly with semiconductor companies.

Demonstrated advantage of wireless communication is shown in the production line. The effect of cabling removal between the battery modules dramatically improves the production time of the battery pack. Additionally, a further optimized design of the modules, the housing and the battery packs can be proposed. For instance, this strategy supports the further penetration of the Cell2Pack technologies in battery pack configuration.

Besides the better packaging of the BMS, wireless communications enable the relocation of the decision layer features, as well as ease the scalability and modularity of the battery, enabling only through software or parametrization update an additional freedom grade of battery designing. Therefore, this step is almost a prerequisite for enabling IoT-based cloud applications like digital twin or battery passport [54]–[61]

In Table III the most investigated wireless architecture approaches are presented:

TABLE III
ANALYSIS OF WIRELESS COMMUNICATION CHARACTERISTICS

Hardware	Bandwidth	Application	Distance
IEEE 802.15.4 - Zigbee	Low speed	Internal	Short –mid range
802.15.4e Smart Mesh	medium speed	External	Mid-high range
802.11ah WiBaAN	Medium speed	External	Mid-high range
802.11 Wlan	Medium speed	External	Mid-high range
IoT / 5G	High speed	External	High range

E. Software Over The Air (SOTA) - Cybersecurity

Batteries are evolving into IoT solutions to use the benefits of cloud systems, but at the same time, it is increasing cybersecurity risks. As explained in [62], the vulnerability of battery systems in electric vehicles to cyberattacks, which can result in various negative consequences, is high. Cyberattacks on the BMS can be classified into the following:

- A DoS attack can flood the BMS controller with requests, leading to delays in responses or even preventing the BMS from sending response messages, which can cause a loss of control and the system's status.

- Tampering with the battery or BMS can lead to catastrophic events going out of SOA.
- Malicious code injection can enable an attacker to gain control over the BMS and affect the system's normal functioning. The worst case could be by doing an unauthorized firmware update, leading the cells and the battery uncontrolled with a high possibility of operating out of SOA.
- Spoofing, replaying, and man-in-the-middle attacks can result in false information being displayed to the control systems, affecting the use of the battery.

The ISO/SAE 21434 [63] and UL 4600 [64] standards provide valuable guidance for the cybersecurity of the BMS but are not sufficient on their own to address all possible cybersecurity risks and threats. It has to be followed by methodologies like secured by design strategies and it should rely on cybersecurity knowledge in other fields such as flow control, source authentication, rate limiting, intrusion detection, remote attestation, and integrity verification.

F. Enabling 2nd life with battery passport in digital twin

The introduction of wireless communication interface either with a centralized BMU governing all the modules with MMU / CMU, or directly communicating with MMU/CMUs do enable the translation of the decision layer outside of the battery pack and its placement in a digital twin running in a cloud platform.

A digital twin consists of a set of software-based models representing a virtual copy, named twin, of a physical hardware, which can have close interaction. This digital twin representation revises, extends or adds features to the existing physical hardware, enabling the use of advanced algorithms, review its parametrization or allows global supervision; features which would not be possible in the physical hardware to be implemented (e.g. virtual sensors) [65]–[68].

Benefits are the introduction of more computational power, enabling more accurate and precise battery models, and state estimation algorithms, which can expand the monitoring capabilities of the BMS using, for instance, virtual sensors. Additional advantage is the introduction of fleet management and statistical approach analysis of the vehicle and battery data, and with the help of data mining tool chains and AI algorithms, a better parametrization dataset can be roll out with less effort to the field with updates applied and distributed using SOTA.

One of the major challenges derived from the introduction of the digital twin and externalized methods of calculation abstracted of the application inside the vehicle is the need of a robust and safe cybersecurity standard application to protect the BMS from receiving an incorrect software update or a parametrization dataset from the cloud. New updates shall be monitored, and

their plausibility analyzed by the monitoring layer before they can be applied and actively driving the BMS.

Once a standard platform for digital twins is established, the introduction of a dynamic Battery Passport (one that updates battery data periodically going beyond the current static data approaches) will require less effort. Most discussion points regarding battery passport currently are the amount and type of data that it should contain, its ownership (OEM, TIERS, final customer, handler) and all security aspects in terms of privacy and cybersecurity [69]–[77]

IV. IMCU – INTELLIGENT MODULE CONTROL UNIT

The module management unit is the electronics system in charge of monitoring the battery module. The creation of an advanced modular battery requires splitting the decision and monitoring layers increasing the computational resources needs on low levels, like the sensors data processing at the module level. The result is a flexible platform able to be used as a collector of measurements but also provide calculations on the battery SOA and even some SoX.

During the execution of the COBRA project a novel electronic device called iMCU has been developed. This device's mission is to gather information from the battery, execute different algorithms and send the information to be analyzed. It has the ability to operate in stand-alone mode (single module configuration), performing all the BMS related tasks, if necessary, or in combination with other iMCUs or a central BMU for larger batteries. The iMCU comprises two electronic PCBs interconnected in piggyback design concept, the core and the peripherals.

- iMCU core, as a generic and reusable PCB, including the microcontroller Aurix TriCore TC397 [78], the low-level power supply circuitry for the microcontroller, basic supervision LEDs and the interfaces towards JTAG debugging, microUSB and towards peripheral board.
- iMCU peripherals, specific and reusable design PCB depending on the application. On COBRA project, several communication interfaces as analog / digital inputs outputs are provided.

Both boards are interconnected, easing the concept of update of peripherals board in a 2nd application, as well as adapting the resources need to a different size of microcontroller, depending on the needs of the specific application, as depicted in next Fig. 4.

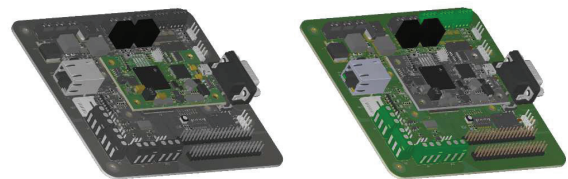


Fig. 4. 3D view of the positioning of the assembly of the core (Top) and peripherals (Bottom) in piggyback connection

A. Technical Description

The intelligence on the iMCU core board is the microcontroller Aurix Tricore TC397 [78]. This device is provided by Infineon, and this specific family of processors combines performance with a robust safety architecture, which makes it the ideal fit for research applications. The microcontroller [79] is a hexacore 32bit RISC microcontroller 300MHz with fully integrated DSP and Floating-point unit capabilities. The iMCU core has three interfaces:

- JTAG to program and debug the firmware in a low level to allow access in any case.
- MicroUSB to program and debug the firmware with an easy method, less devices are needed but less detail in the debugging is obtained.
- The interface between the iMCU core and iMCU peripherals aims to pass all the capabilities of the core board to the peripherals. It is an interface of 180 pins to connect the iMCU peripherals.

The iMCU peripheral's board purpose is to adapt all the functionalities provided by the iMCU core to the specifications and needs of the battery. This board can be adapted to fulfil the needs of any battery research project or type of battery. In the case of COBRA project, the peripherals board has a specific design to fit the project specific needs. The peripherals board is in charge of acquiring information from the cells. In this particular case, smart sensors with different wireless communication channels are in use, consequently the board incorporates a gateway interfaced by two UART ports and a third channel with ISO UART for wired back-up communications. The board has a microSD holder to store information and log data. It uses an external ADC to integrate another type of sensors that provide analogic values, like a current sensor. There are several general-purpose inputs and outputs (GPIOs) to provide flexibility for future applications. In addition, the board has an isolated CAN bus and an Ethernet connector to transfer the information to external devices. In Fig. 5, there are depicted the iMCU boards with each functionality and communication module highlighted.

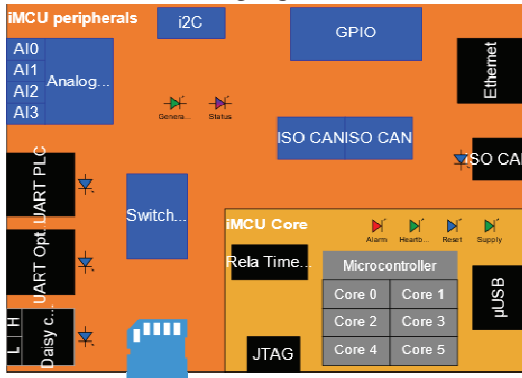


Fig. 5. iMCU components at core and peripherals boards

B. Sensors and new feature implementation

On the iMCU, apart from the regular temperature and cell voltage measurements, it can read new sensors with analogue or digital signals. The new sensors included in this development are:

The strain gauge to measure the deformation of the cells, useful for the type of pouch cell used in the project [41], [80].

Pressure sensor and gas sensor to detect any anomaly at the module and pack level.

Cell-level EIS to evaluate their status deeply.

The information gathered by these sensors is then used to calculate battery state and SoX parameters [81]–[83]. The smart sensor at cell level incorporate the balancing circuit which is triggered by the iMCU.

In Table IV a relationship of the improvement of the decision layer algorithms based on the additional sensor data is summarized. These new physical engineering units are plausibilized by the monitoring layer and received from the sensing layer, therefore, the algorithms used for decision taking are already trusting on the inputs received, as they are crosschecked for errors and validated [41], [84]–[86].

TABLE IV
IMPROVEMENT OF FEATURE CONTENT OF BMS DUE TO ENHANCED SENSING

Sensor type	Feature Relationship							
	Charge Ctrl	Thermal Mgt	Thermal Run.	SOC	SOH	SOF	SOE	SOP
Pressure			X					X
Gas			X					X
EIS	X		X	X	X	X	X	X
Strain Gauge					X	X	X	X
Electrode Volt.	X				X		X	X
Internal Temperature	X	X	X					X
Humidity			X					X

C. Flexibility and extension capabilities

The iMCU is adaptable to the needs of different batteries and projects thanks to its modular nature. They can be combined and re-arranged in several configurations, its software and data exchange protocols are oriented towards multi-unit operation, with or without a centralized battery controller. The core board and its basic software do not require adaptation, while the peripherals board can be changed to fulfil the new requirements in terms of battery type, new sensors, new communications protocols, or other changes that may be required. Also, the computational power of the microcontroller allows the execution of highly demanding algorithms such as Artificial Intelligence [87] to execute trained models.

D. Research purpose

The approach presented in this paper is not aiming at an industrial solution for a cost-oriented product, it serves as an open platform for research purposes. For this reason, once it is finished, the schematics and partial codes will be published and shared as open source to share this platform with the research community. Therefore, KPIs typically defined for an automotive application or cost focused environment are not considered in this paper in detail. Future works can be more oriented to perform a profound technical-economic analysis of the costs introduced per additional feature included in the BMS design aiming to optimize it.

A qualitative and comparative analysis of the current BMS technology of the proposed advancements versus state-of-the-art BMS is discussed in following Table V.

TABLE V
QUALITATIVE EVALUATION OF KPIS OF BMS PROPOSED
(O SIMILAR, + BENEFIT, - DRAWBACK)

	Performance	Cost	Complexity	Scalability
Sensors				
Pressure	+	-	-	-
Gas	+	-	--	-
EIS	++	--	--	++
Strain Gauge	+	--	--	++
Electrode Volt.	+	--	--	++
Internal Temperature	+	--	-	++
Humidity	+	-	-	-
Communication				
CAN	o	+	o	-
Daisy Chain, isoUART, IsoSPI	o	o	o	o
Wireless	-	-	+	+
Actuators				
Passive Balancing	o	o	o	o
Active Balancing	++	-	-	+
Reconfiguration	++	--	--	++

V. CURRENT IMPLEMENTATION

As COBRA project implements a smart sensor unit at cell level, cell measurements and balancing are placed directly in such external component and communicating with iMCU wirelessly and/or through iso UART. Such physical values (cell voltage, cell temperature, strain gauge, EIS) are not implemented in the first version of peripheral board although a pressure sensor is integrated here and directly wired to peripherals board.

The software under development is also designed to be modular (as the hardware) but considering the limitations that the microcontroller and the C programming language bring in this regard.

Each functionality and communication type has been developed as separate functions and then passed as references to the main program to have a main program to manage the different tasks. Due to this separation, it is possible to create some external, precompiled files, to include functions developed by others without the need to

share the code inside these functions.

In terms of communications and to be useful for new developments is planned to use standard file descriptors of communications, like CAN DBC, to change the protocol or the variables position of the communication quickly.

At the current time, the iMCU is in the manufacturing process. Fig. 6 shows a picture of the boards with the distribution of the connectors.

The next step is to perform the first validation tests and evaluate the prototype in COBRA's battery.

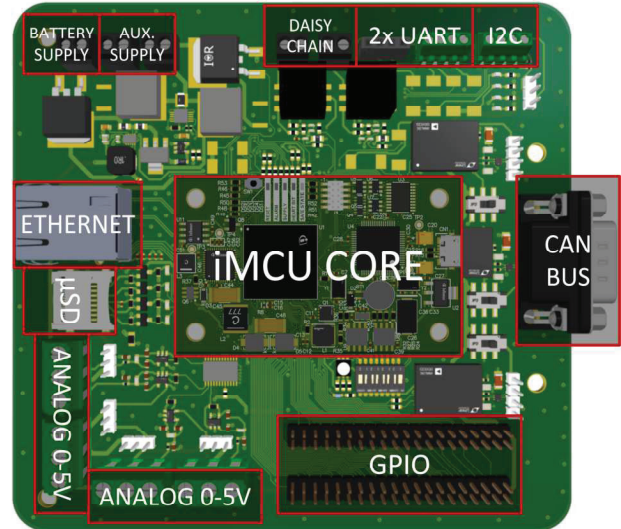


Fig. 6. iMCU boards capture of the current implementation.

VI. CONCLUSION

In this work a study and analysis of the trends on BMS developments (sensing technologies, active balancing, reconfiguration, wireless communications, cybersecurity requirements and 2nd life needs) for smart battery management systems is provided.

Observing the studies, a structured analysis of the features impacted is performed and an iMCU research platform based in a 2 boards approach is discussed and designed.

The next steps are to set into operation the first version of the iMCU board and set the basis for the future development of additional peripherals / core boards for consequent focused analysis of the trends identified.

After first validation with a real battery, additional designs are intended to be produced and validated as well in real environment for research and teaching purposes.

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